

Eissa Islam

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years building production-grade systems across LLMs, RAG pipelines, computer vision, and MLOps. Specialize in end-to-end delivery — from architecture design to cloud deployment — on real projects at a university, fintech, and government initiatives. Published IEEE author (ITC-Egypt 2024). Proven track record in agentic AI workflows, autonomous agents, conversational AI, and data-intensive applications that solve tangible problems.

EXPERIENCE

AI Engineer — Misr University for Science & Technology · Giza, Egypt *Feb 2025 – Present*

- Spearheading 3 production AI systems on campus: RoboMUST (voice-enabled RAG learning platform), ShopIntel (LLM-powered shopping assistant with live scraping), and a YOLOv8 garbage classification system for smart recycling.
- Deployed a campus-wide facial recognition attendance system automating record-keeping across multiple university departments.

AI Engineer — AiTech · Egypt *Oct 2024 – Dec 2024*

- Built an intelligent RAG chat system (web scraping + MongoDB + ChromaDB) and a Quranic chatbot grounded in classical tafsir using a dual-vector semantic search pipeline.
- Developed ScrapeMaster, a Tkinter desktop tool for automated YouTube metadata and image scraping used for internal dataset collection.

ML Engineer Intern — Digital Egypt Initiative (MCIT) · Egypt *Apr 2024 – Oct 2024*

- Applied MLOps (MLflow, Hugging Face, Azure AI) on fraud detection and NLP pipeline projects, delivering deployment-ready models through structured evaluation workflows.

Data Scientist Trainee — Banque Misr — Rowad Internship · Egypt *Jun 2024 – Aug 2024*

- Built fraud detection and credit scoring models on imbalanced datasets using SMOTE and ensemble methods, collaborating with analysts in an Agile environment.

Data Science Trainee — ALX Africa & ExploreAI · Remote *May 2023 – Aug 2024*

- Completed capstone projects in model training, evaluation, and dashboard development using deep learning techniques on real-world datasets.

ML Intern — Misr University AI Tech R&D Center · Giza, Egypt *Jan 2023 – Jul 2023*

- Developed a TensorFlow/Keras deep learning model for prostate cancer detection, leading all phases from DICOM preprocessing to final validation.

EDUCATION

B.Sc. Information Technology — Major: Artificial Intelligence *Sept 2019 – June 2024*

Misr University for Science and Technology, Giza, Egypt | GPA: 3.41 / 4.0

KEY PROJECTS

JobPilot AI — Autonomous Job Application Agent *Tech: Python 3.11, Scraping, Playwright, Gemini API, Groq/Llama 3, SQLite, SQLAlchemy, Alembic, Streamlit, python-telegram-bot, Jinja2, python-docx, FPDF2, APScheduler, Docling, Oracle Cloud, GitHub Actions, Sentry*

- Built a fully autonomous 7-layer Python system scraping job postings 24/7 across 4 platforms using anti-bot stealth scraping, with a hybrid scoring engine (rule-based + LLM semantic analysis) rating each job 0–100 against a candidate profile.
- Implemented an AI-powered CV tailoring module generating ATS-optimized DOCX/PDF files per job via prompt-engineered LLM calls — producing 80+ tailored CVs — with a Telegram bot for human-in-the-loop approval before submission.
- Deployed on Oracle Cloud Always-Free ARM VM at zero cost via systemd + GitHub Actions CI/CD, with a Streamlit dashboard featuring a Skills Gap Analyzer and automated PDF weekly reports.

Enterprise Brain — AI-Powered Enterprise Knowledge & Operations System *Tech: Python, LangGraph, FastAPI, BGE-m3, Qdrant, Celery, Redis, Ollama, Docling, Webhooks, Google Drive API, Slack API, React/Next.js, SSE*

- Architected a multi-agent RAG system (Router, Retrieval, and Action agents via LangGraph) that ingests enterprise data from Google Drive, Slack, Notion, and Jira in real-time via Webhooks + Celery/Redis workers, enabling employees to query and act on company knowledge through a single interface.
- Engineered a multi-modal ingestion pipeline (text, Excel, OCR images, audio) with BGE-m3 embeddings and semantic re-ranking, plus enterprise-grade RBAC ensuring users only access vectors permitted by their role.
- Delivered a React dashboard with real-time SSE streaming, clickable source citations, and human-in-the-loop approval gates for AI-proposed actions.

RoboMUST — AI-Powered Educational Platform *Tech: Python, FastAPI, HTML/CSS/JS, MongoDB, ChromaDB, LangChain, Ollama, Whisper, TTS, RAG, Docling*

- Built and deployed a full-stack voice and text AI platform at Misr University enabling students and professors to interact with course material via RAG pipelines, quiz generation, and smart summarization using local LLMs.
- Delivered end-to-end solo: AI logic, FastAPI backend, MongoDB/ChromaDB storage, REST APIs, and complete web interface.

Graduation Project: Parkinsonian Gait Severity Assessment (IEEE Published) *Tech: Python, Mediapipe, Scikit-learn, Keras, Pandas, NumPy*

- Led development of a vision-based gait analysis system for dementia patients achieving 92% validation accuracy using 2D/3D video data and a multi-model ML ensemble.
- Published at IEEE ITC-Egypt 2024 — IEEE Xplore

TECHNICAL SKILLS

Languages & DBs	Python, SQL, MongoDB, ChromaDB, PostgreSQL, SQLite
ML / DL	TensorFlow, PyTorch, Scikit-learn, Keras, Hugging Face Transformers
LLM & RAG	LangChain, LangGraph, Ollama, Gemini API, Groq (Llama 3), BGE-m3, Semantic Search, Re-ranking, Prompt Engineering
MLOps & Cloud	MLflow, Azure AI, AWS, Oracle Cloud (ARM), GitHub Actions, Sentry, systemd
Web & APIs	FastAPI, Flask, Streamlit, REST APIs, SSE, Uvicorn, HTML/CSS/JS
Scraping & Auto	Scrapling, Selenium, Playwright, BeautifulSoup, httpx, n8n, APScheduler, Docling, Celery, Redis
Data & Viz	Pandas, NumPy, Matplotlib, Plotly, Power BI, Excel
Vector DBs	ChromaDB, Qdrant, Milvus, PGVector
Computer Vision	YOLOv8, OpenCV, Mediapipe, Object Detection, Pose Estimation
Soft Skills	Agile / Scrum, Technical Leadership, Research, Problem-solving

PUBLICATION

Parkinsonian Gait Severity in Older Adults with Dementia Using Natural Gait Video Analysis
IEEE 2024 — International Telecommunications Conference (ITC-Egypt) View on IEEE Xplore